Torque-speed and efficiency report

Validation SPM 24p — Small. Built on demand from the design's flux-linkage map and the drive limits set on the Torque-speed tab, with the same calculation the tab uses. The run's own report records the single solved operating point.

Key results

Peak torque	1.468 N m
Base speed	5935.428 rpm
Peak power	0.93 kW
Top speed reached	7679.646 rpm
Best efficiency	92.602%

Status

Qualified outputs pass a benchmark gate against a reference; experimental outputs have none yet.

Torque-speed envelope	Experimental: Not yet compared with measured torque-speed data. Interpolated from the flux-linkage map with sinusoidal current; inverter drops and dead time are ignored.
Efficiency map	Experimental: Not yet compared with measured efficiency data. MTPA control: the smallest current for each torque within the voltage limit, interpolated from the flux-linkage map; torque is electromagnetic. Copper loss at the winding temperature; iron loss scaled from the map's reference speed, hysteresis with frequency and eddy current with its square.
Map flux linkage, MTPA trajectory and inductances	Experimental: The d/q gates cover 12 operating points, not a map's full range of currents and angles, and do not bound the inductances.
Phase resistance and copper loss	Experimental: No wire is set, so the wire is sized to fill the slot to its limit: an estimate, not the resistance of a real winding. Set the wire for a resistance the phase-resistance gate covers.

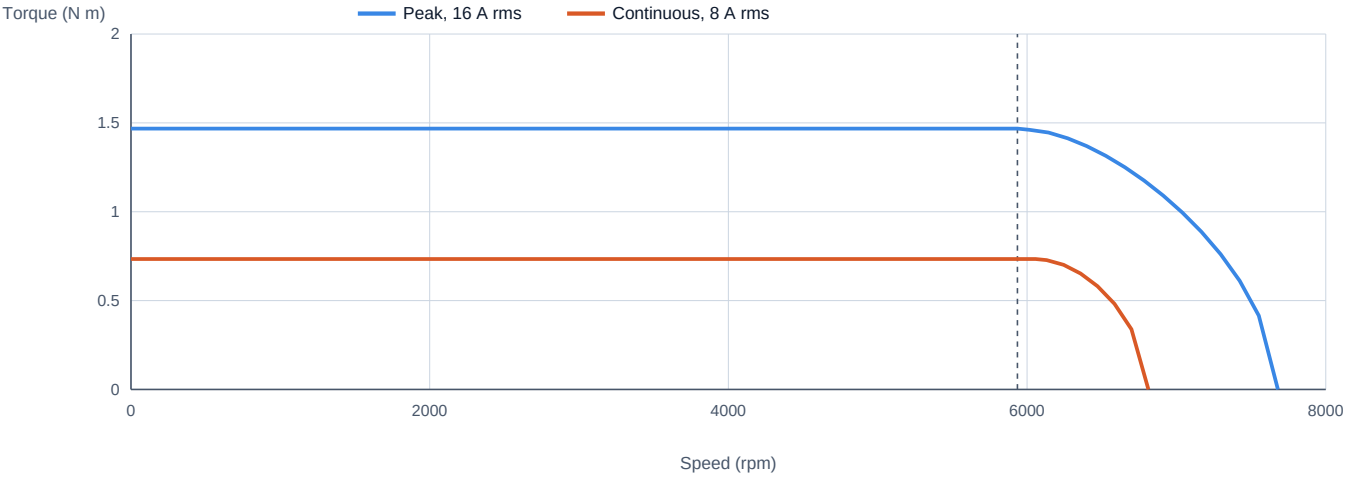
Drive inputs

Peak current	16 A rms (22.627 A peak)
Continuous current	8 A rms (11.314 A peak)
DC bus	48 V, as entered
Phase voltage limit	27.713 V peak (modulation index 1.155)
Top speed	18000 rpm
Drive control	MTPA, then field weakening (least current per torque)
Phase resistance	0.014 ohm at 20 degC, experimental
Mechanical loss	0 W
Flux-linkage map	7 currents up to 22.627 A peak x 10 angles (0 to 90 deg), 24 rotor positions each, normal mesh, standard grid; iron loss at 6000 rpm
Map scope key	d8cd037c42569b9e381a3e9b93760fad93daf950c0ce4ea5b97033d86eb9104f
coilEM	3ce46ed1ba1bcf19b422e504248917843dcf22cf

Torque and power against speed

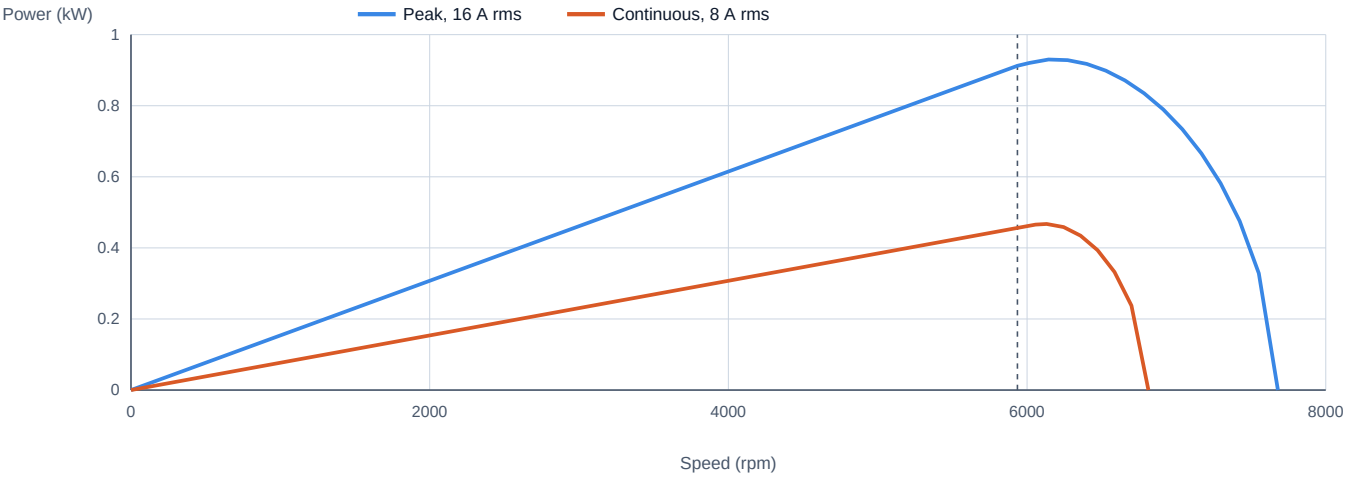
Torque against speed

Experimental



Power against speed

Experimental



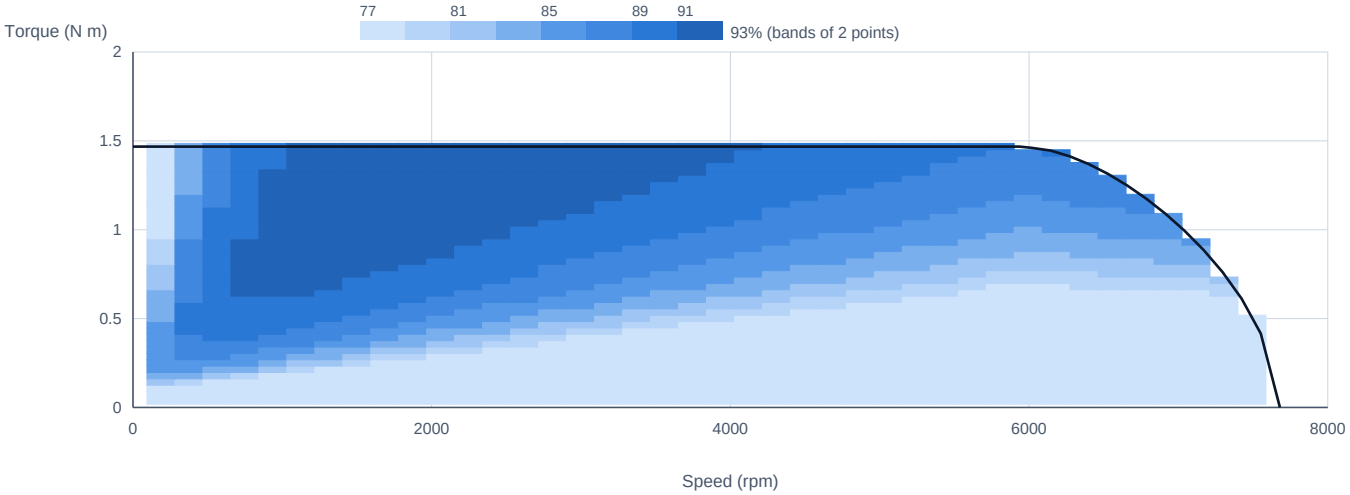
Peak curve at selected speeds (every sample is in the performance CSV)

Speed (rpm)	Torque (N m)	Power (kW)	Current (A rms)	Angle (deg)	Voltage (V peak)	Region
0	1.468	0	16	0	0.309	MTPA
767.965	1.468	0.118	16	0	3.85	MTPA
1407.935	1.468	0.216	16	0	6.805	MTPA
2175.9	1.468	0.334	16	0	10.351	MTPA
2815.87	1.468	0.433	16	0	13.307	MTPA
3583.835	1.468	0.551	16	0	16.853	MTPA
4223.806	1.468	0.649	16	0	19.808	MTPA
4991.77	1.468	0.767	16	0	23.355	MTPA
5631.741	1.468	0.866	16	0	26.31	MTPA
6271.711	1.413	0.928	16	15.585	27.713	Field weakening
6911.682	1.091	0.79	16	41.941	27.713	Field weakening
7679.646	5.35e-6	4.302e-6	16	90	27.713	Field weakening

Efficiency map

Efficiency map, MTPA, then field weakening

Experimental

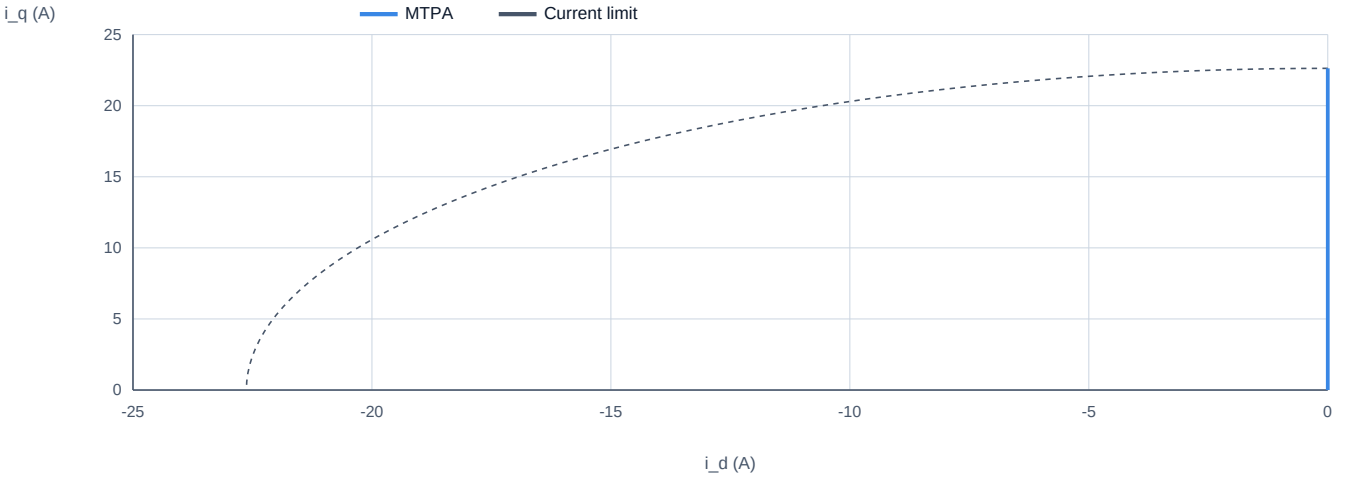


Efficiency = shaft power / input power, with shaft power = T x omega - iron loss - mechanical loss and input power = T x omega + copper loss. Cells above the peak curve are outside the drive's reach. The current and angle of every cell are in the performance CSV.

MTPA trajectory and flux linkage

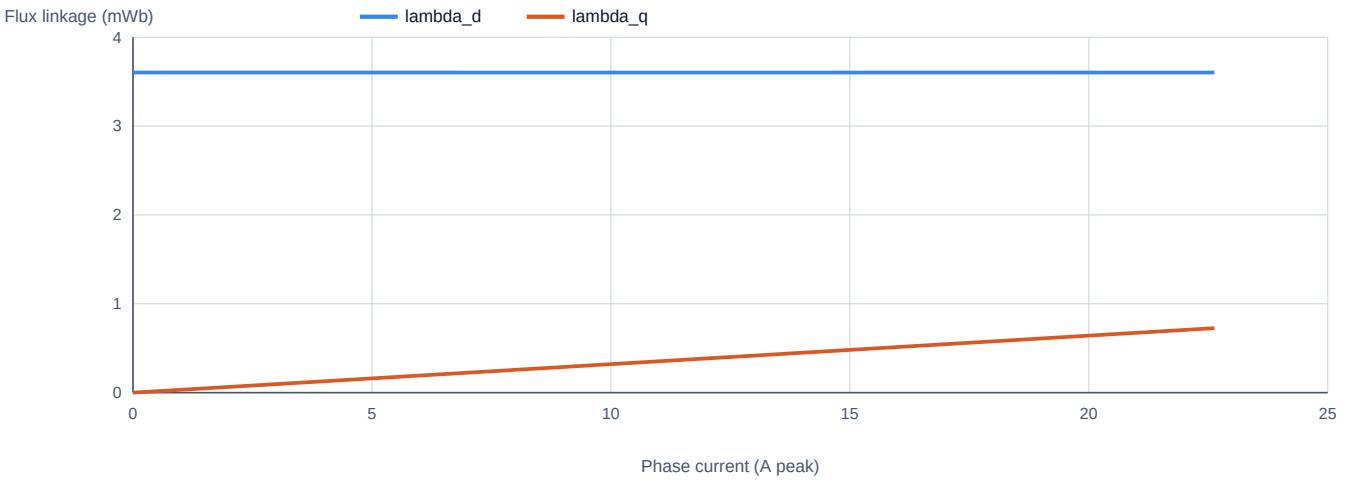
MTPA trajectory in the id-iq plane (peak A)

Experimental



Flux linkage along the MTPA trajectory

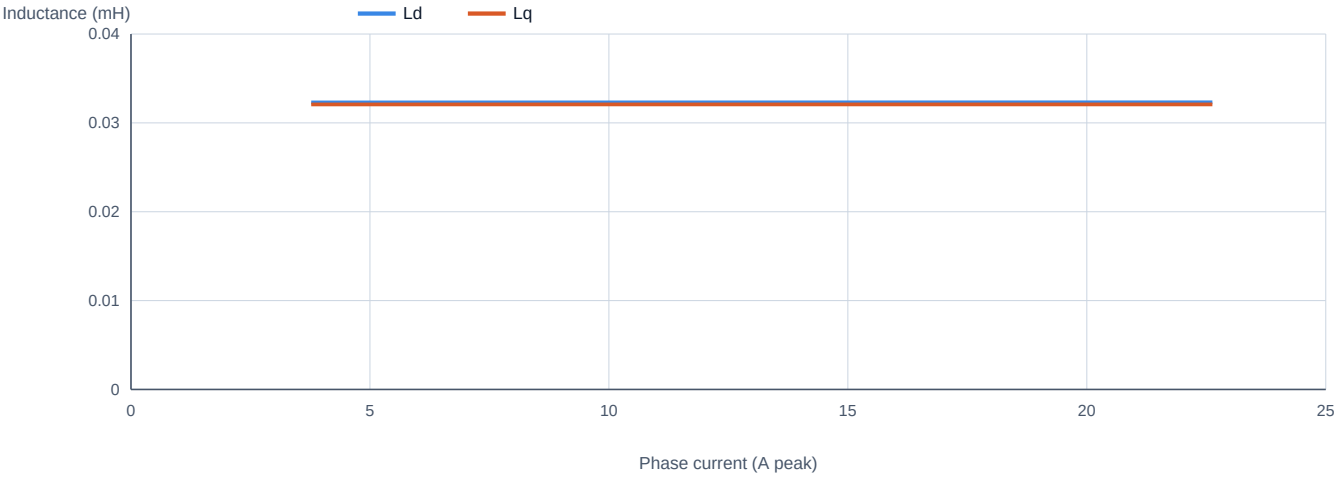
Experimental



Apparent inductance

Apparent inductance against current

Experimental



Current (A peak)	Current (A rms)	Ld (mH)	Lq (mH)
3.771	2.667	0.032	0.032
7.542	5.333	0.032	0.032
11.314	8	0.032	0.032
15.085	10.667	0.032	0.032
18.856	13.333	0.032	0.032
22.627	16	0.032	0.032

Notes

- FEM torque is 100.0-100.1% of the d/q torque across 53 map points.

- Ld from the 90 deg points, (lambda_d - lambda_pm) / i_d; Lq from the 0 deg points, lambda_q / i_q. Both are secant values and include saturation.

- The MTPA angle is read off the interpolated map, whose angles are 10 deg apart at most. At low current, where torque hardly changes with angle, it is approximate; a finer map grid gives a smoother trajectory.

- Not yet compared with measured torque-speed data. Interpolated from the flux-linkage map with sinusoidal current; inverter drops and dead time are ignored.

- Not yet compared with measured efficiency data. MTPA control: the smallest current for each torque within the voltage limit, interpolated from the flux-linkage map; torque is electromagnetic. Copper loss at the winding temperature; iron loss scaled from the map's reference speed, hysteresis with frequency and eddy current with its square.

Known limitations

- Interpolated from a flux-linkage map (Catmull-Rom over current and current angle); nothing is extrapolated beyond its currents and angles. Not yet compared with measured data.

- Steady state with sinusoidal current. Inverter voltage drops, dead time and the voltage behind the iron-loss resistance are left out.

- Torque is electromagnetic. Shaft output subtracts iron loss and the mechanical loss entered for the design.

- Iron loss is scaled from the map's reference speed: hysteresis with frequency, eddy current with its square.